

basic radio

Cross-channel connectivity. Ability to share or exchange information between diverse systems

Software radio design

Radio design requires a broad set of design skills. A higher skill level is needed for almost all aspects of the radio design because of the interdependency of the radio sub-systems.

It is important to ensure that characteristics like flexibility, complete and easy reconfigurability, and scalability are present in the final product. The generic design procedure for software radios (shown in Fig. 3) follows and demonstrates the interaction between the various sub-systems of the radio design:

Step 1 (systems engineering). Allocation of sufficient resources to establish the service, given the system's constraints and requirements.

Step 2 (RF chain planning). Simultaneous flexibility in selection of power gain, bandwidth, centre frequency, sensitivity and dynamic range. Achieving strict flexibility is impractical and trade-offs must be made.

Step 3 (analogue-to-digital and digital-to-analogue conversion selection). This selection requires trading power consumption, dynamic range and sample rate. Analogue-to-digital conversion and digital-to-analogue conversion selection is closely tied to the RF requirements for dynamic range and frequency translation. It is the weakest link in the overall design.

Step 4 (software architecture selection). It ensures expansion, compatibility and scalability for the software radio. Ideally, the architecture should allow for hardware independence through the appropriate use of middleware (interface between software and the hardware layer).

Step 5 (digital signal processing hardware architecture selection). Digital signal processing can be implemented through DSPs, FPGAs and/or ASICs. Typically, DSPs offer maximum flexibility, lowest power consumption and highest computational rate.

FPGAs, on the other hand, lie somewhere between ASICs and DSPs in these characteristics.

Step 6 (radio validation). It is essential to ensure that the system is failproof and that the communicating units operate correctly. Testing and validation steps can be taken to help minimise the risk.

Driving factors

The factors driving the wider acceptance of software radio are ease of design, ease of manufacture, ease of upgrades, multifunctionality, compactness, power efficiency, fewer discrete components and use of advanced signal processing.

Ease of design. The time required to develop a marketable product is a key consideration in modern engineering design. Software radio implementation reduces the design cycles for new products, freeing the designers from the hard work associated with analogue hardware designs.

Ease of manufacture. RF components are hard to standardise, and may have varying performance characteristics. Optimisation of the components in terms of performance may take a significant amount of time and thereby delay product introduction. In general, digitisation of the signal early in the receiver chain can result in a design that incorporates a few discrete parts, resulting in reduced inventory for the manufacturer.

Ease of upgrades. In the course of deployment, current services may need to be updated or new services may have to be introduced. Such enhancements have to be made without disrupting the operations of the current system. A flexible architecture like the SDR allows for improvements and additional functionality without the expense of replacing all the old units.

Multi-functionality. With the development of short-range services like Bluetooth and IEEE 802.11, it is now possible to enhance the services of a radio by leveraging other devices that offer complementary services. The re-configuration capability of software

radio can support an almost infinite variety of services in a single system.

Compactness and power efficiency. The SDR approach results in a compact and power-efficient design; as the number of systems increases, the same piece of hardware is reused to implement multiple systems and interfaces.

Fewer discrete components. A single high-speed digital processor may be able to implement many traditional radio functions such as synchronisation, demodulation, modulation, error correction, source coding, encryption and decryption, thereby reducing the number of required components and the size and cost of the radio.

Use of advanced signal processing. The availability of high-speed signal processing on board the radio allows implementation of new receiver structures and signal processing techniques. Techniques such as adaptive equalisation, interference rejection, channel estimation, anti-jamming, cross-polarisation, interference cancellation, maximum likelihood algorithms and strong encryption—previously deemed too complex—are now implemented on high-performance DSPs, as part of the SDR platform.

Mobile communication: A promising application

Mobile communication requires high-performance signal processing technique to allow operation as close as possible to the Shannon information theoretic bound. However, not only must these systems provide exceptional performance, but due to the market and fiscal pressure, they must be flexible enough to allow rapid tracking of the evolving standards.

Software-defined radios are emerging as a viable solution for meeting the conflicting demands in mobile communication. These support multimode and multiband modes of operation, allowing service providers an economic means of future-proofing their increasingly complex and costly systems. ●

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